Test-beam topology and response-model studies of scintillator range staves for HIBEAM/NNBAR

HIBEAM/NNBAR CCB test-beam working group
Working publication package

12 August 2026

Working publication – not submission-authorising. This PDF is the canonical chapterised build surface for the paper, but the Cycle-3 adversarial audit on `main@32fd72b675aa4cba0da3bdee52561755be960ada` invalidates promotion of the current ΔE–E production numbers, the July optical calibration grid, and the 8.9% deposited-energy-resolution headline. Those quantities are kept only in clearly marked publication-hold text until their physics/provenance issues close. The authoritative status is tracked in the repository claim ledger and publication audit files.

Contents

Abstract

The HIBEAM/NNBAR programme is developing an annihilation detector for neutron-conversion searches, including free neutron–antineutron oscillation. One proposed detector element is a range stack of plastic-scintillator staves intended to sample the stopping pattern and energy loss of low-energy charged particles. This paper studies the CCB test-beam range-stack prototype using beam data and Geant4-based detector simulations.

The analysed B-stack data contain four readout channels labelled B2, B4, B6 and B8, while the corresponding detector model contains eight physical longitudinal layers. The beam data show a strong difference in longitudinal occupancy between the declared Sample-I and Sample-II run families: one selected population is concentrated near the stack entrance while the other populates downstream channels more frequently. Because primary run-log and hardware-trigger records remain incomplete, the publication-safe statement is a *run-family-dependent stopping/penetration topology*; a quantitative trigger-efficiency interpretation remains gated.

The physically relevant amplitude telescope observable is $\Delta E = A(B2)$ and $E = A(B4) + A(B6) + A(B8)$. Sparse longitudinal readout can alias the Bragg-rise region when the highest stopping power occurs in an uninstrumented layer. The current numerical ΔE–E data/MC production bundle is retained under publication hold pending regeneration from the authorising waveform schema, a source-bound layer map, corrected full-layer MC aggregation, and a production MC sample with a valid source-measure/provenance contract.

The located beam waveform product contains eight channels with 16 samples per channel. A direct B4–B6 timing analysis yields an approximately 38 ns central residual width, but the waveform window, component assignment and timing reference do not permit extraction of an intrinsic stave resolution. The result is therefore reported only as a format-limited negative timing result.

A detailed single-stave optical model provides a framework for separating quenching, scintillation, wavelength shifting, optical transport, SiPM response and electronics. The July optical calibration grid and its derived held-out energy-reconstruction result are now classified as superseded model diagnostics because the campaign predates source-level optical fixes and because its energy variable was Birks-visible energy rather than unquenched deposited energy. The paper therefore reports the detector-response methodology and the evidence needed for quantitative closure, while withholding absolute light yield and detector energy-resolution claims until the regenerated model and systematic programme are complete.

1 Introduction

The HIBEAM/NNBAR programme at the European Spallation Source is designed to search for neutron-conversion processes, including neutron–antineutron oscillation. A free neutron–antineutron transition would violate baryon number by two units and would probe physics beyond the Standard Model [1]. The annihilation detector must identify the products of an antineutron interaction in a thin target and reconstruct a low-energy hadronic final state containing pions, nuclear fragments and secondary interactions.

The detector operates under constraints that differ from those of a collider spectrometer. The neutron flight region must remain magnetically quiet, limiting conventional magnetic momentum measurement. Many charged annihilation products also lie below the energy range in which a standard high-energy sampling calorimeter is most effective. HIBEAM/NNBAR detector studies therefore proposed a range measurement based on multiple layers of plastic scintillator followed by an electromagnetic calorimeter [2]. The range stack can provide the depth reached by a charged particle, the pattern of energy loss along its path and an inter-layer timing observable.

The CCB test beam was used to study this range-stack concept with charged particles whose stopping behaviour can be modelled in Geant4. This paper asks five questions:

(i) what longitudinal stopping/penetration differences are measured between the declared beam-data run families;

(ii) how much information is lost when only alternating B-stack layers appear in the analysed readout;

(iii) what timing statement is supported by the authorising waveform product;

(iv) what causal response stages must be included in a one-fibre, one-end optical model; and

(v) how should deposited/visible-energy reconstruction be defined and validated without converting simulation assumptions into detector calibration constants.

Four evidence classes are used throughout. **Beam-data measurement** denotes a quantity computed from source-bound CCB data for a declared population. **Truth-level MC** denotes Geant4 particle or energy-deposit information before detector response. **Model-dependent detector MC** denotes a numerical prediction after optical/electronics modelling when important response parameters remain hypotheses. **Calibrated detector result** is reserved for a response whose relevant nuisance parameters and data lineage have independent evidence and held-out closure.

> **Evidence boundary.** This chapter deliberately separates measured data, simulation truth and model-dependent response. The distinction is part of the scientific result: a reproducible simulation number is not automatically a detector-performance measurement.

2 CCB test-beam configuration

Hardware facts, simulation-configuration values and unresolved legacy narratives are separated in the publication hardware bill of materials. Each quantity should carry one of four statuses: MEASURED, DESIGN_SPEC, SIM_CONFIG, or UNKNOWN_EXTERNAL. Simulation agreement is not hardware provenance.

2.1 Beam, target and two-arm layout

The reviewed CCB simulation configuration uses a 190 MeV incident proton beam and a deuterated target. The model contains the elastic channel

$$p + d \to p + d,$$

and transports the outgoing charged particles toward two scintillator arms. Historical source configuration uses a nominal geometry-distance parameter of 109 cm, with the B stack near $-38°$ and the A stack near $+71.5°$, and contains eight B bars and four A bars. These values are publication-safe as simulation/configuration quantities; they are not survey-grade metrology until matched to primary collaboration records.

The compact detector simulation contains the target, scintillator stack volumes, trigger-related volumes and ProtoTPC components. Source review shows that it is useful for first-order acceptance and truth-energy-deposit studies but does not yet represent every passive material relevant to precision stopping predictions. Wrapping, supports, photosensor boards, cabling, dead layers and survey uncertainties must therefore be treated explicitly when the paper makes material-budget claims.

Figure not promoted into the publication build

`figures/illustrative/geometry_tof.png`

ILLUSTRATIVE; source-bound survey figure still required.

Figure 1: CCB two-arm geometry schematic. The current package treats this as illustrative/configuration evidence, not survey metrology.

2.2 B-stack readout and run families

The analysed data use four B-stack channels labelled B2, B4, B6 and B8. The simulation contains eight physical B layers, so the data represent alternating longitudinal samples. The exact offset between detector labels and Geant4 copy/layer IDs remains a publication blocker: historical issue #869 argued for layers 1/3/5/7 while later paper/BOM material uses 0/2/4/6. Only the every-other-layer structure is currently robust. A final data-matched MC figure must import a single source-bound detector map rather than hard-code either convention.

The current analysis configuration separates Sample-I and Sample-II run families and also separates calibration and analysis roles. However, the exact hardware trigger thresholds, coincidence logic, prescales and run-purpose record remain partly external. The safest beam-data language is therefore *Sample-I/Sample-II run-family comparison*. The MC selection based on first-stack-layer charged hits is retained only as `MC_TRIGGER_PROXY` until the hardware response is modelled or demonstrated equivalent.

> **PUBLICATION HOLD.** Do not use the words "trigger efficiency", "hardware-trigger reproduction", or quantitative trigger-induced species fractions until issues #962, #1045 and #1296 close on the same run/hardware truth surface.

3 Scintillator stave and readout

3.1 Stave geometry

The current design specification follows the later collaboration clarification associated with issue #796. Each B stave is described as an extruded-polystyrene scintillator approximately 50 cm long, 5.18 cm wide and 2.0 cm thick along the nominal particle path. Two longitudinal holes of 2.0 mm diameter are separated by 2.0 cm centre-to-centre and accept 1.8 mm Kuraray Y-11 wavelength-shifting fibres. The exterior includes a reflective coating. Older repository prose describing approximately one-metre BC-408 bars is treated as superseded/unknown external history rather than an alternative current specification.

The B-stave concept supports two fibres read from two ends, corresponding to four possible optical measurements. The beam-test readout used only one fibre at one end according to the collaboration design clarification. This one-ended configuration makes response dependent on longitudinal hit position and removes the charge/time asymmetry that a two-ended system could use to constrain position.

Figure not promoted into the publication build

`figures/final/stave_geometry.pdf`

PENDING hardware/BOM closure; see issue #1296.

Figure 2: Target publication figure for the stave geometry. It remains a placeholder until the hardware drawing/BOM is source-bound.

3.2 SiPM and electronics boundary

The detector model uses a Hamamatsu S13360-3050CS MPPC. Manufacturer data specify a $3\,\mathrm{mm}\times 3\,\mathrm{mm}$ photosensitive area, $50\ \mu$m pixel pitch and 3600 pixels, with peak sensitivity near the Y-11 green-emission band [7]. These public specifications do not by themselves determine the CCB operating response. PDE, overvoltage, temperature, optical coupling, illumination footprint, microcell recovery, crosstalk, afterpulsing, gain and the analogue impulse response are separate response objects.

The publication model therefore keeps sensor photons, primary avalanches, correlated avalanches, charge and ADC waveform observables distinct. A scalar efficiency is not allowed to absorb unresolved physics from multiple stages.

3.3 Waveform products

Two waveform schemas appear in the project history. Historical reconstruction configurations use 18 samples per channel, whereas the located LUNARC beam product contains eight channels with 16 samples per channel (128 waveform words per event). The provenance analysis under issue #993 classifies these as distinct schemas rather than a proven reversible 16-to-18 sample transformation. Consequently, historical sub-nanosecond timing studies based on the 18-sample product cannot be promoted to the authorising 8-by-16 beam sample.

Every publication amplitude/timing figure must record waveform schema, raw-file hashes, producer revision, channel map and pulse-polarity contract. The publication source package intentionally does not hide these provenance requirements behind a generic label such as “raw data”.

4 Simulation programmes

4.1 CCB event simulation

The CCB event simulation provides particle identities, trajectories and energy deposits across the two-arm detector. Historical production used Geant4 electromagnetic and hadronic/ion transport suitable as a starting point for proton/deuteron studies in this energy domain [4, 5]. The simulation is used for truth-level stopping and for testing how detector segmentation changes observables.

The publication audit identified important source-model constraints. Legacy samples can contain non-unit `PrimaryWeight` values from a generator source-measure implementation whose frame/Jacobian semantics were later found to be defective. The exact paper MC file `output_krakow_1M.root` is byte-hashed but is not yet bound to a complete production receipt containing generator revision, cross-section table/mode, seed, physics list, build/toolchain and all geometry/material configuration. Until issue #1311 closes, that file is a historical provenance-gated diagnostic rather than the nominal publication MC.

4.2 Single-stave optical simulation

The single-stave model transports the charged particle through a detailed scintillator/fibre geometry and follows optical photons through scintillation, wavelength shifting, attenuation and sensor boundaries. A separate SiPM/electronics layer converts sensor arrivals into avalanche/charge/waveform quantities.

The model is scientifically most useful when represented as a causal graph:

$$E_{\mathrm{raw}} \to E_{\mathrm{vis}} \to N_{\gamma,\mathrm{scint}} \to N_{\gamma,\mathrm{WLS\,abs}} \to N_{\gamma,\mathrm{WLS\,emit}} \to N_{\gamma,\mathrm{sensor}} \to N_{\mathrm{primary\,aval}} \to Q \to A_{\mathrm{ADC}}.$$

Here E_{raw} is unquenched deposited energy and E_{vis} is the Birks-visible energy. These two variables must never share the same bare E_{dep} label.

4.3 Simulation evidence classes

Paper figures distinguish `TRUTH_LEVEL_MC`, `DIGITISED_MC`, `MODEL_DEPENDENT_OPTICAL_MC`, and `CALIBRATED_DETECTOR_MC`. The last category is intentionally empty until response nuisances and held-out data closure satisfy the publication gate.

5 Data taking and event selections

5.1 Run families

Current configuration files define Sample-I and Sample-II analysis populations and separate calibration populations. These configuration roles are useful for reproducible processing but are not a substitute for the primary hardware run log. The run-role discrepancy discussed under issue #962 means that calibration/analysis assignment must be bound to the exact fitted object before timing or response calibration can be described as held out.

The historical S00 selected-pulse product contains 640,737 B-stack pulses. That count is deterministic for the fixed historical producer/input set, but the canonical claim ledger gates publication promotion because the raw waveform width, raw-to-sorted lineage and channel-polarity/hardware contracts are not all independently closed. Selected-pulse counts are not detector efficiencies.

5.2 Measured longitudinal topology

The robust data-side observation is that the declared Sample-I and Sample-II populations have different longitudinal B-stack occupancy patterns: one population is concentrated near the entrance while the other has a larger downstream fraction. This is the central measured range-stack result that survives the current audit.

Figure not promoted into the publication build

`figures/final/data_depth_profile.pdf`

PENDING regeneration from the authorising 8-by-16 event-level amplitude product.

Figure 3: Target beam-data B-stack depth profile for Sample-I and Sample-II.

Evidence boundary. Until the hardware run/trigger record closes, the paper should call this a run-family-dependent longitudinal topology. Interpreting the difference uniquely as a trigger effect would confound trigger changes with possible run-period beam/electronics changes.

5.3 Monte Carlo population comparison

The corresponding MC population is currently formed with `MC_TRIGGER_PROXY`. It can show that the transport model contains physically plausible stopping/penetrating populations, but it cannot authorise a hardware-trigger efficiency or a precise proton/deuteron mixture until event-level species counting, trigger response, source measure and production provenance all close.

Figure not promoted into the publication build

`figures/final/mc_depth_profile.pdf`

PENDING source-measure, trigger-response, event-unit and MC-production provenance closure.

Figure 4: Target weighted MC depth and species-composition figure matched to the final data population definitions.

6 Inter-stave timing

6.1 Observable and covariance

For staves i and j, define

$$\Delta t_{ij} = t_j - t_i - t_{\mathrm{TOF},ij}.$$

The pair width does not uniquely determine single-stave resolutions. In general,

$$\mathrm{Var}(\Delta t_{ij}) = \sigma_i^2 + \sigma_j^2 - 2\,\mathrm{Cov}(t_i, t_j).$$

Common trigger phase, sampling phase, reconstruction choices and event conditions may produce covariance. The paper therefore does not divide a pair width by $\sqrt{2}$ without an independently justified model.

6.2 Pulse-component definition

Leading-edge timing is amplitude-dependent. Constant-fraction timing also becomes ambiguous for multi-component waveforms if the threshold is defined from the global maximum: changing the fraction can switch the timed physical component. The production method must define the selected pulse component before optimising any timing fraction. Same-sample minimum widths across a fraction scan are exploratory model-selection diagnostics, not performance measurements.

6.3 Authorising data result

On the located 8-by-16 raw waveform product, a direct B4–B6 CFD analysis yields 5,207 events with valid times and an approximately 38 ns central-68% residual width after the applied TOF subtraction. Peak positions are multi-modal and frequently lie near waveform boundaries.

> **Evidence boundary.** The 38 ns width is reported as a **format-/reconstruction-limited residual**. The nominal 10 ns sampling interval alone is not sufficient to explain a 38 ns width; window phase, pulse-component ambiguity, boundary truncation and timing-reference/TOF assumptions remain entangled. No intrinsic stave timing resolution is extracted.

Figure not promoted into the publication build

`figures/final/timing_b4_b6_residual.pdf`

PENDING final figure regeneration; caption must state NOT DETECTOR RESOLUTION.

Figure 5: Target B4–B6 residual figure from the authorising 8-by-16 waveform schema.

Historical sub-nanosecond values are retained only as provenance history or simulation/method-closure diagnostics and must not appear in the abstract/conclusions as beam-data performance.

7 Amplitude ΔE–E and the segmentation limit

7.1 Observable definition

The supervisor-defined beam-data analogue of a range-telescope observable is

$$\Delta E_{\text{data}} = A(B2), \qquad E_{\text{data}} = A(B4) + A(B6) + A(B8),$$

where A is a baseline-subtracted ADC amplitude. These axes are response proxies, not calibrated energies.

A data-matched MC view should use the same four physical readout positions,

$$\Delta E_{\text{MC},4} = E_{\text{dep}}(B2), \qquad E_{\text{MC},4} = E_{\text{dep}}(B4) + E_{\text{dep}}(B6) + E_{\text{dep}}(B8),$$

while a full-segmentation truth view must sum each unique downstream physical layer exactly once. A B2-versus-B4 plot is a two-channel response diagnostic and must not be called ΔE–E.

7.2 Sparse-segmentation mechanism

Alternating readout positions can miss the layer in which the stopping-power rise becomes largest. A change in entry energy, angle or passive material can move the Bragg-rise region between sampled and unsampled layers and thereby alter the apparent sparse ΔE–E topology. The publication test is therefore a controlled comparison of full physical layers and the exact data-matched readout mask, with readout phase treated as a nuisance until the hardware map is source-bound.

7.3 Cycle-3 status of the current production bundle

> **PUBLICATION HOLD.** The numerical bundle currently stored under `reports/paper_956_deltaE_E_20260812T103800Z/` is **not** publication-authorising. The Cycle-3 audit found that (i) its beam-data input is the historical 18-sample selected-pulse product despite the 8-by-16 authorising raw schema; (ii) downstream amplitudes are threshold-censored before missing channels are zero-filled; (iii) the MC builder aliases readout columns onto physical-layer names and corrupts the supposed full-downstream sum; (iv) the exact even/odd layer-map offset is unresolved; (v) the 7000-ADC "saturation" boundary is not source-bound hardware saturation; and (vi) the legacy MC source measure/production receipt is gated.

The quarantined plots are kept under `publication/figures/gated/` to preserve provenance without allowing accidental use in the paper.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/fig07_data_deltaE_E_per_sample.png`

GATED by Cycle-3 audit; do not cite as final result.

Figure 6: Quarantined current beam-data amplitude ΔE–E plot.

The final Figure 7/8 production must start from an event-level table that contains all four reconstructed channel amplitudes before analysis-threshold censoring, then apply declared threshold scans; import the source-bound readout map; keep physical-layer and readout namespaces distinct; and use a production MC whose source measure and run provenance are authorising.

8 Monte Carlo prediction of light collection and transport

8.1 Stage-resolved response

The optical model should report separately the production of scintillation photons, WLS absorption, WLS re-emission, guided/transported photons, sensor arrivals, primary SiPM avalanches, correlated avalanches, charge and ADC response. Useful conditional efficiencies include

$$\epsilon_{\mathrm{WLS\,capture}} = \frac{N_{\gamma,\mathrm{WLS\,abs}}}{N_{\gamma,\mathrm{scint}}}, \quad \epsilon_{\mathrm{transport}} = \frac{N_{\gamma,\mathrm{sensor}}}{N_{\gamma,\mathrm{WLS\,emit}}}, \quad \epsilon_{\mathrm{PDE}} = \frac{N_{\mathrm{primary\,aval}}}{N_{\gamma,\mathrm{sensor}}}.$$

These ratios should be studied versus hit position, species/deposit regime and model nuisance parameters.

8.2 Historical July calibration grid

The July single-stave grid contains five proton/deuteron points and historically produced an order-ten detected-PE per visible-MeV scale. It is useful as a regression/history object, but it is not the nominal publication model.

> **PUBLICATION HOLD.** The exact July calibration revision predates multiple source-level fixes. In particular, inner and outer fibre claddings both used the shared NIST `G4_PLEXIGLASS` material and attached different refractive-index tables, allowing the later assignment to overwrite the intended inner-clad optical properties. The campaign also predates explicit WLS fluorescence-yield/multiplicity, separated scintillator/Y-11 attenuation controls, corrected reflector-surface semantics, source-bound fibre-to-SiPM interface modelling and a single canonical SiPM response state. Its historical "PE/MeV" denominator is the Birks-visible variable, not unquenched deposited energy. Issue #1303 therefore requires regeneration before any numerical light-yield result is promoted.

The historical values and plots remain available under `publication/figures/gated/` and the source report for correction history only.

8.3 Required nominal optical campaign

A publication campaign must bind a clean source revision and build receipt; preserve E_{raw} and E_{vis} separately; record stage counters; use the physical one-fibre/one-end channel with other ends only as controls; scan or state the support in longitudinal/transverse position and angle; and propagate independent high-impact optical nuisance families rather than re-tuning the total mean response back to a historical target.

9 Energy reconstruction and resolution

9.1 Measurand definition

Energy resolution is not a single detector-independent number. For the single-stave response chain, two physically distinct targets are available:

$$E_{\text{raw}} \equiv \text{unquenched Geant4 energy deposited in the scintillator,} \tag{1}$$
$$E_{\text{vis}} \equiv \text{Birks-visible/quenched energy used to generate scintillation light.} \tag{2}$$

A full range-stack reconstruction may later target incident charged-particle kinetic energy from amplitudes and stopping depth. These estimands must not share one bare E_{dep} label.

For a chosen target E_T, define

$$r = \frac{E_{\text{reco}} - E_T}{E_T}.$$

A publication result should report median bias, σ_{68}, RMS, tail fraction and uncertainty intervals on held-out populations, as functions of target energy and relevant phase-space variables.

9.2 Status of the current held-out result

> **PUBLICATION HOLD.** The current 8.9% held-out σ_{68} result is not a publication deposited-energy resolution. The producer reconstructs `edep_scint_MeV`, which is the Birks-visible variable, while the manuscript/issue called it Geant4 deposited energy. The saturation diagnostic was sign-reversed and clipped to zero, executed-script provenance was incomplete, position transfer

was not tested, and the two held-out grid points occupy nearly the same target-energy regime. In addition, the underlying optical grid is superseded by issue #1303. The plot is therefore quarantined pending a regenerated optical campaign and an explicit E_{raw} versus E_{vis} target choice.

GATED FIGURE – NOT RENDERED IN PUBLICATION PDF

`figures/gated/edep_reconstruction_heldout.png`

GATED: wrong/ambiguous energy semantics and superseded optical campaign.

Figure 7: Quarantined Figure 11 from the historical held-out reconstruction study.

9.3 Required publication analysis

The replacement study should freeze train/calibration and evaluation populations before fitting; include more than one effective target-energy regime; validate particle-species transfer without truth-label tuning; test longitudinal/transverse position and angle support; compare transparent linear/monotonic baselines before any ML model; and propagate optical/SiPM model nuisance families separately from finite-simulation statistics. If a resolution parametrisation is fitted, the available energy grid must identify its terms.

A full-stack incident-energy estimator may use the amplitude vector $\mathbf{A} = (A_{B2}, A_{B4}, A_{B6}, A_{B8})$ and stopping depth, but it remains downstream of material-budget, trigger/selection and per-channel response closure.

10 Discussion

The CCB beam data demonstrate that the scintillator stack resolves markedly different longitudinal selected populations. This is the strongest direct experimental observation currently available for the paper. Its interpretation as a trigger effect is plausible and consistent with the intended run families, but the publication text should remain at run-family level until the hardware trigger/run log is source-bound.

Sparse longitudinal sampling is a central design lesson. An alternating four-channel readout can retain stopping-depth information while losing the local sequence of energy loss near the Bragg rise. This creates a real aliasing mechanism: changes in energy, angle or passive material can move the high-dE/dx region across the readout phase. A full detector should therefore instrument sufficient longitudinal positions, or otherwise constrain the missing layer response, before using a sparse ΔE–E band as a high-precision PID observable.

The timing analysis gives a complementary negative result. The located waveform schema is adequate for amplitude and coarse timing diagnostics but not for an intrinsic sub-nanosecond stave-resolution measurement. This result is scientifically useful because it states exactly what must be recorded or calibrated in a future campaign: sufficient waveform window, source-bound sampling/timebase, stable component assignment and an independent timing reference/covariance model.

The optical-model audit shows why a detector-response paper must distinguish a causal response graph from a tuned total efficiency. WLS fluorescence yield, trapping/attenuation, end coupling, PDE, microcell occupancy/recovery, correlated noise and electronics can compensate one another in the mean. Stage-resolved counters, phase-space scans and independent bench/hardware constraints are more informative than matching one PE/ADC number.

The current publication package intentionally treats invalidated results as *gated objects* rather than deleting them. This preserves provenance and makes reviewer-visible the difference between a historical result, a reproducible diagnostic and an authorising measurement.

11 Conclusions

The CCB test-beam programme provides a measured demonstration that different selected run families populate different depths of a scintillator range stack. The analysed B readout samples only alternating longitudinal positions, making the response sensitive to where the stopping-power rise falls relative to instrumented layers. The correct telescope-style amplitude observable is $\Delta E = A(B2)$ and $E = A(B4) + A(B6) + A(B8)$; a B2–B4 plot is only a two-channel diagnostic.

The located 8-by-16 waveform product supports a negative timing conclusion rather than a detector-resolution number: the B4–B6 residual is broad and strongly affected by the finite waveform representation and component-selection problem. Historical sub-nanosecond results are therefore not transferred to the authorising beam sample.

A single-stave Geant4 optical model remains valuable as a response framework, but the historical July calibration campaign is superseded as the nominal model and must be regenerated after source-level optical fixes. Likewise, the current held-out energy reconstruction is quarantined because its target was Birks-visible energy rather than the raw deposited-energy quantity claimed in the manuscript, and because its underlying optical campaign is superseded.

The paper is therefore not yet a quantitative detector-performance publication. The path to submission is well defined: source-bind the hardware/run/channel map; reconstruct authorising event-level beam amplitudes; regenerate the ΔE–E comparison with a valid production MC; regenerate the optical grid with explicit energy semantics and response stages; perform a broader held-out energy-reconstruction/systematic study; then propagate every accepted result through the canonical claim ledger, source tables, figures and manuscript before a final adversarial review.

12 Reproducibility and figure-generation requirements

Every quantitative paper figure must be generated from a tracked machine-readable result or source table. The producing record must include source commit, producer hash, input paths and SHA-256 values, event/run population, selection, weights, uncertainty method and evidence class. No headline number should be manually copied into a plotting script or caption.

Data and MC must use compatible observable definitions. Beam ADC amplitudes and truth-level Geant4 energy deposits may be shown in separate panels, but they must not be placed on a shared calibrated energy axis without a validated response transformation. Weighted MC results must state the weight semantics and report $\sum w$, $\sum w^2$, effective sample size and weight pathologies relevant to the estimand.

Timing and energy calibrations must be fit on declared calibration/training populations and evaluated on held-out runs or simulation populations without retuning. Resampling must preserve the physical/statistical source unit, normally DAQ run/event for beam data and generator event for MC, rather than assuming every exported hit/pulse row is independent.

The canonical publication workflow is

$$\begin{aligned}\text{physics/provenance contract} &\rightarrow \text{result + manifest + source table}\\ &\rightarrow \text{claim ledger} \rightarrow \text{figure/table}\\ &\rightarrow \text{manuscript} \rightarrow \text{adversarial review.}\end{aligned}$$

An issue is not publication-complete when code alone runs successfully.

A Publication status and open gates

This appendix is part of the working PDF so that exported copies cannot lose the scientific readiness state.

Gate	Publication consequence
#1296	Hardware/BOM/run record: source-bind installed stave, channel map, target, trigger and survey/run settings.
#962 / #1045	Run-role and trigger response: prevents over-interpreting Sample-I/Sample-II differences as pure trigger effects.
#869 / #954	Readout-map and waveform polarity: required before authorising event-level amplitude observables.
#956	Regenerate beam/MC ΔE–E from the authorising schema and corrected MC layer namespaces.
#1178 / #1179 / #1311	MC source measure, uncertainty and production provenance.
#1088 / #1303	WLS photophysics and regeneration of the single-stave optical grid after superseded source defects.
#1302 / #1297	Raw deposited energy vs Birks-visible energy semantics and a corrected held-out reconstruction study.
#1304	Single scientific source of truth for claim status across figures, WIKI and manuscript.
#1301 / #1305	Umbrella submission and merged-state fail-closed gates.

The Cycle-3 audit documents the detailed failure modes under `chatgpt_todo/PAPER_SUBMISSION_AUDIT_CYCLE3_20260812.md` and its addenda.

B Primary repository evidence surfaces

The working paper is derived from, and must remain synchronized with, the following repository evidence surfaces:

- `docs/claim_ledger.csv` – canonical scientific claim status;
- `paper/hardware_bom.csv` – publication hardware truth/status surface;
- `paper/figures.yaml` – historical figure registry, currently subject to governance repair under #1304;
- `chatgpt_todo/PAPER_SUBMISSION_AUDIT_CYCLE3_20260812.md` and addenda – current fail-closed review;

- `reports/studies/data_side/REPORT.md` – located raw beam-data audit and timing limitation;
- `reports/1781181864.166832.35d806b2__s21_geant4_source_review/REPORT.md` – source-level CCB Geant4 review;
- `docs/adr/ADR-WLS-FLUORESCENCE-YIELD-UNVERIFIED.md` – absolute WLS-yield boundary;
- `docs/adr/ADR-SIPM-PHYSICS-BLOCKED-WAVEA-LANE01.md` – SiPM-response boundary.

The previous Markdown working manuscript is retained under `publication/source/manuscript_working_20260812.md` for provenance. The chapterised TeX files in this folder are the canonical editing surface going forward.

References

[1] S.-C. Yiu et al., "Status of the Design of an Annihilation Detector to Observe Neutron-Antineutron Conversions at the European Spallation Source," *Symmetry* **14** (2022) 76. doi:10.3390/sym14010076.

[2] K. Dunne et al., "The HIBEAM/NNBAR Calorimeter Prototype," *Journal of Physics: Conference Series* **2374** (2022) 012014. doi:10.1088/1742-6596/2374/1/012014; arXiv:2107.02147.

[3] J. Barrow et al., "A Computing and Detector Simulation Framework for the HIBEAM/NNBAR Experimental Program at the ESS," *EPJ Web of Conferences* **251** (2021) 02062. doi:10.1051/epjconf/202125102062; arXiv:2106.15898.

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[5] J. Allison et al., "Recent developments in Geant4," *Nucl. Instrum. Meth. A* **835** (2016) 186–225. doi:10.1016/j.nima.2016.06.125.

[6] Kuraray Co., Ltd., "Spectral and Characteristic Data: Wavelength Converting Fiber," Y-11(200) technical data. Manufacturer values are representative rather than guaranteed detector-specific specifications.

[7] Hamamatsu Photonics K.K., "MPPC S13360-3050CS," official product specifications.